ENSEMBLE LEARNING

→ Most powerful ML techniques is ensemble learning

→ It is a combination of multiple ML models.

→ It increase the reliability and accuracy of predictions

→ Ensemble learning is a process of training multiple ML models and combining their outputs together

→ Multiple models are used as a base to create one optimal predictive model.

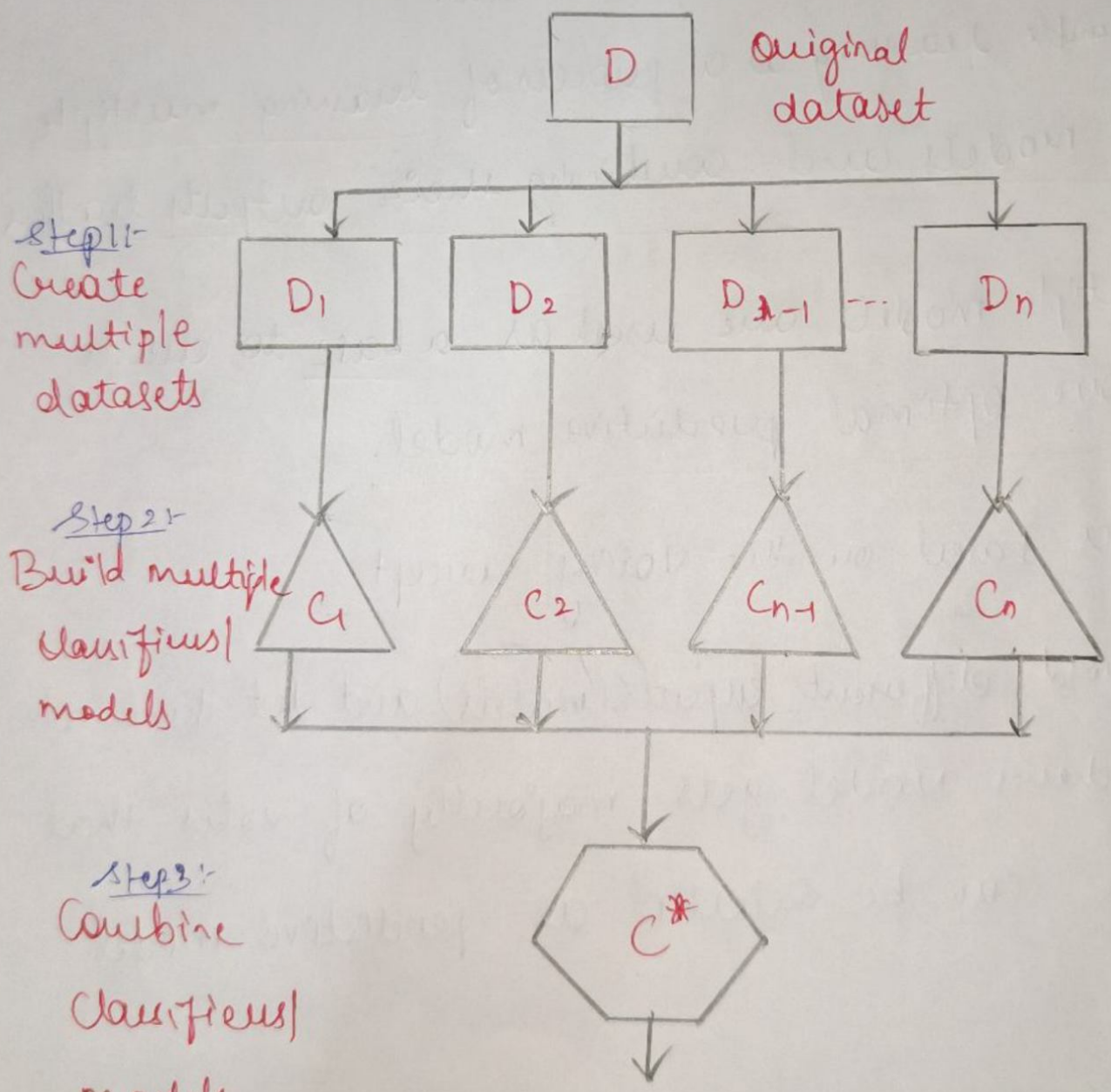
→ It is based on the voting concept.

→ Build different experts (models) and let them vote.

→ Whichever model gets majority of votes that can be selected as predictive model.

→ Here we will be training multiple models with same dataset and by considering all the predictions of model, we will get final output.

GENERAL IDEA :-



1. Given original dataset (D)
2. Create multiple datasets ($D_1, D_2 \ldots D_n$)
3. Using each dataset develop a ML model/ classifiers ($C_1, C_2 \ldots C_n$)
4. Combine the classifiers/models (C^*)
5. Check which model is getting majority votes
6. Select that model as base model
7. Make predictions on unseen data / test data.

ENSEMBLE TECHNIQUES:-

1. Bagging
   - ↳ Random forest
2. Boosting
   - ↳ Adaboost
   - ↳ Gradient boosting
   - ↳ xgboost

① BAGGING :-

→ also called as "bootstrap aggregation"

① Original dataset D.

② Creating multiple base models $M_1, M_2, M_3 \ldots M_n$ using dataset.

③ For each and every model we provide of sample of data/records using original dataset.

④ Again resample the records and give it to model 2. ..

⑤ This is called as "row sampling with Replacement".

⑥ For each and every model we use row sampling with replacement each time when we provide sample of records.

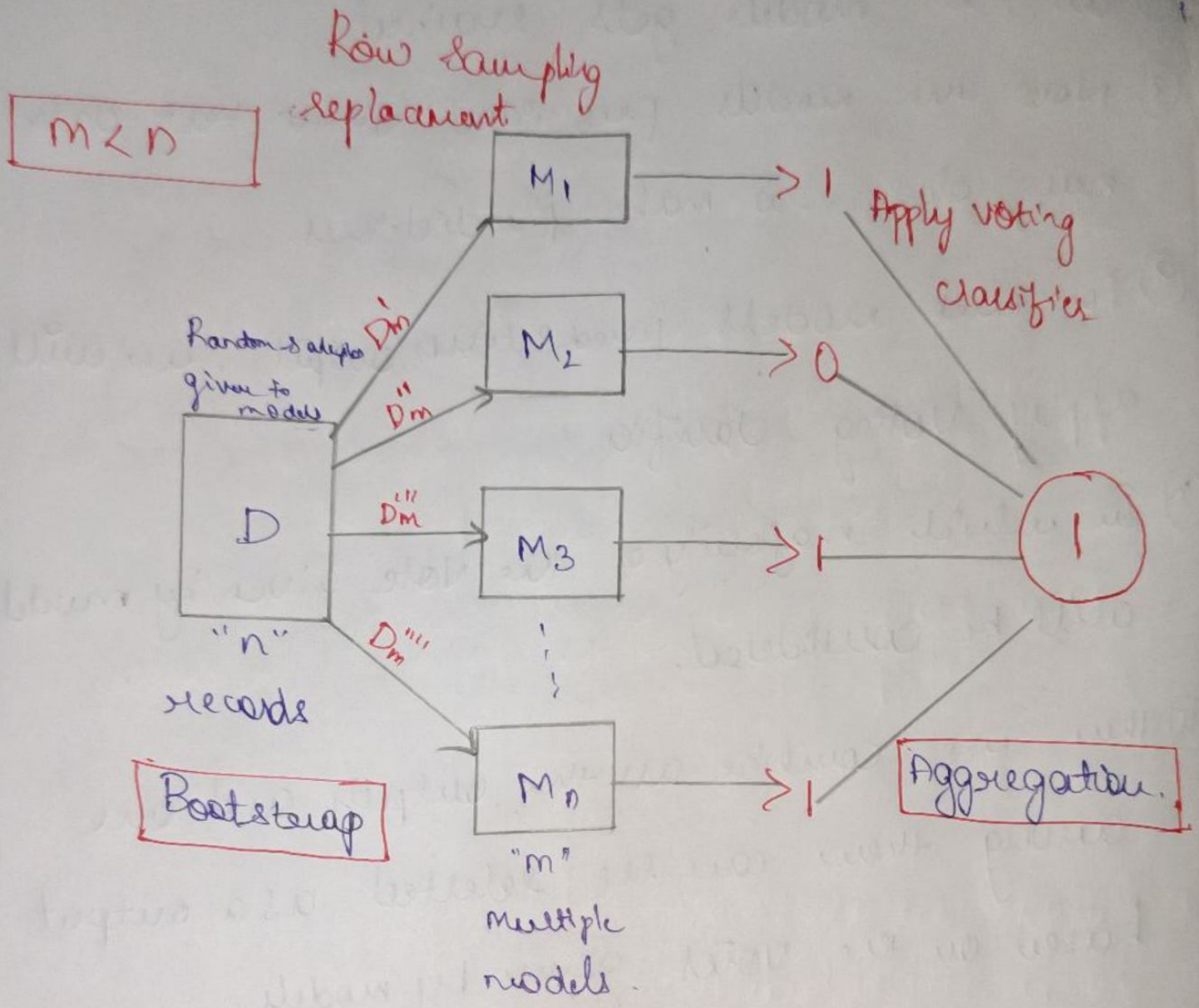
(7) all the models gets trained.

(8) Now the models provided with test data/ new data to make predictions.

(9) For each models predictions/output, we will apply voting classifier.

(10) In which majority of the vote given by model will be considered.

(11) Then we combine all the outputs and one among them can be selected as a output based on the votes given by models.



D → original dataset

D_m' → random samples to model 1

D_m'' → random samples to model 2

D_m''' → random samples to model 3

D_m'''' → random samples to model n.

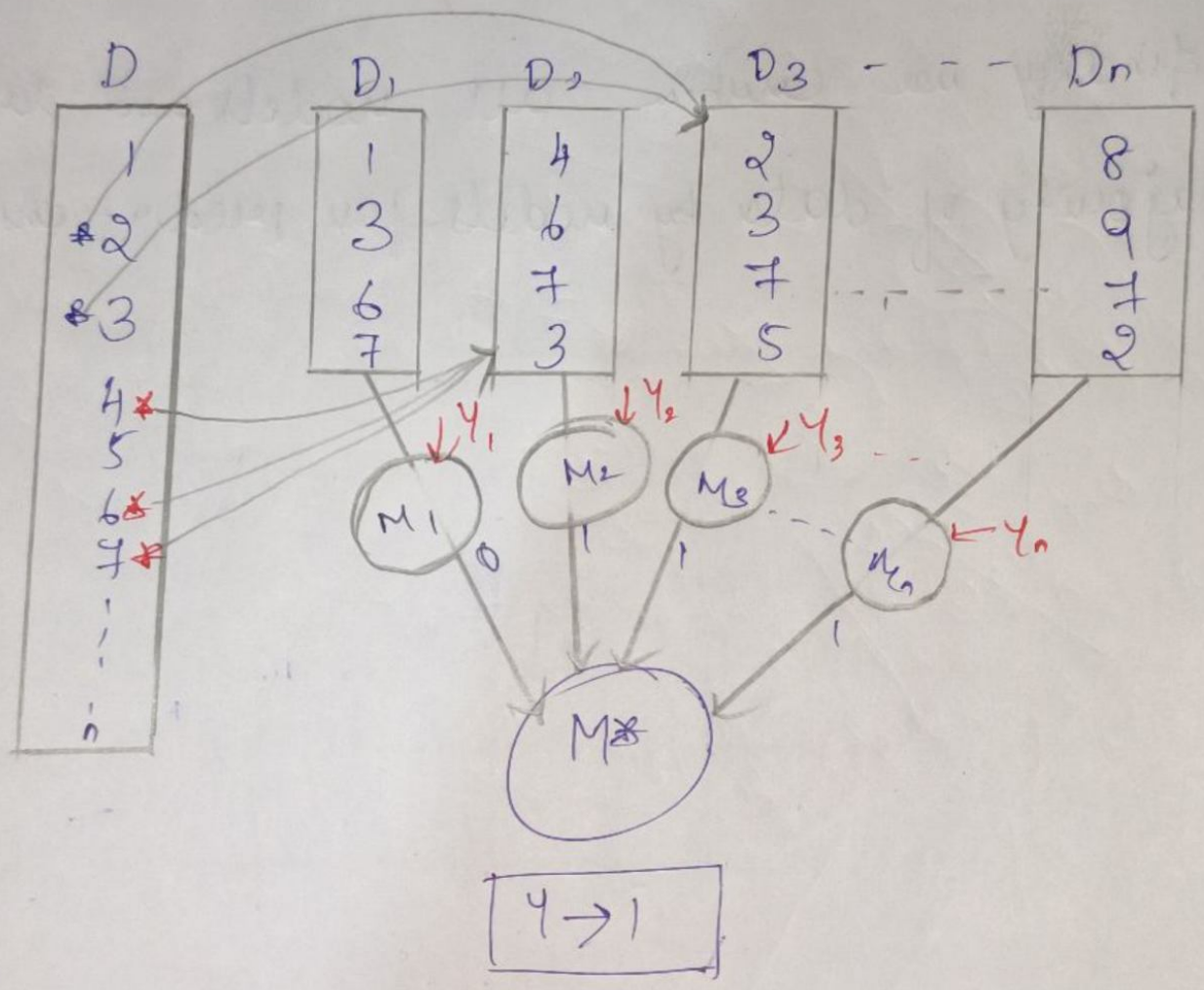
Example for bagging :-

Original data	1	2	3	4	5	6	7	8	9	10
Bagging (Round1)	7	8	10	8	2	5	10	10	5	9
Bagging (Round2)	1	4	9	1	2	3	2	7	3	2
Bagging (Round3)	1	8	5	10	5	5	9	6	3	7

→ Now using above we will construct a classifier for multiple models.

x	1	2	3	4	5	6	7	8	9	10
	1	1	1	-1	-1	-1	-1	1	1	1

② BOOSTING:-



→ Here we have a training data D.

→ Randomly we give data to multiple models M1, M2 -- Mn

→ Suppose M1 wrongly classifies 4, 6, 7, this data will be given to model M2 for training and so on

② BOOSTING (ADABOOST).

→ It is also called as sequence ensemble method

→ It follows some sequence

① Given a original dataset D with training instances.

② We train to create multiple models.

(3) Here in boosting, each training instances/records are assigned with weights ($w_1, w_2, \ldots w_n$)

(4) Initially all the weights have same values.

(5) Let's say we are creating model $D_1, D_2 \ldots D_n$

(6) Using original dataset D, random samples get selected and given to D_1 model for training.

(7) For making the predictions, all the original dataset is given to model D_1.

(8) Model D_1 makes predictions on all the instances of D.

(9) The model D_1, miss classifies some instances of D.

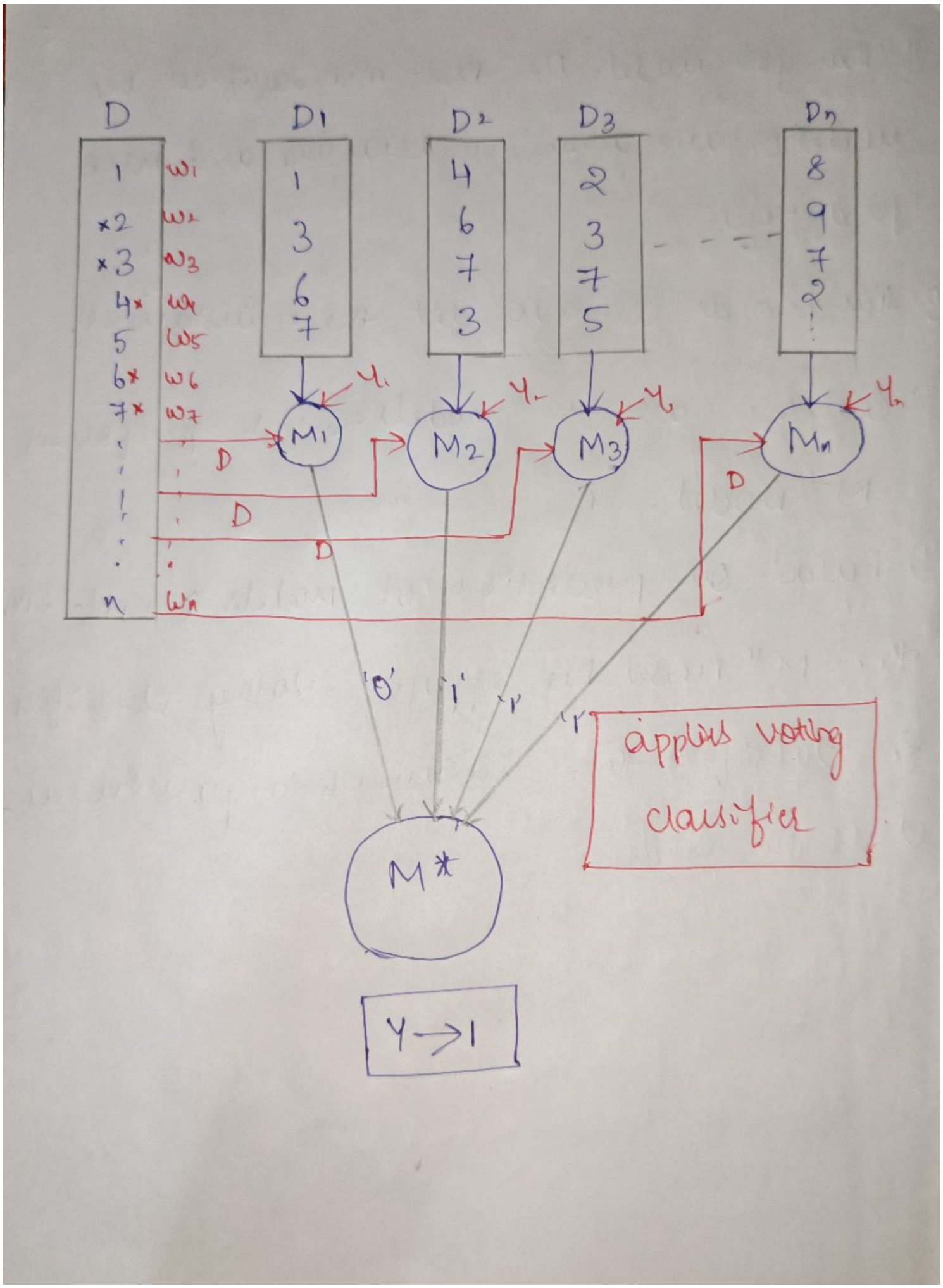
(10) The weights of misclassified instances are updated (increasing the weights).

(11) Then for model D_2, the misclassified by model D_1 are given for training and make predictions.

(12) this process continues till the n instances.

(13) Combine all the models which performs M^* model.

(14) Based on predictions of models $M_1, M_2 - M_n$ the M^* model by applying voting classifier, classifies the instances to particular class or labels.


D
D1
D2
D3
Dn
1
×2
×3
4×
5
6×
7×
n
w1
w2
w3
w4
w5
w6
w7
wn
1
3
6
7
4
6
7
3
2
3
7
5
8
9
7
2
M1
M2
M3
Mn
D
'0'
'1'
applies voting classifier
M*
Y→1